SCIENCE 1.3 **Name:** ***ANSWERS***

WORKSHEET SIX **INHERITANCE**

1. Match up the following terms with their appropriate descriptors.

(a) Genotype	**2**	(1) A gene that carry identical alleles.
(b) Zygote	**7**	(2) The pattern of genes present in cells
(c) Phenotype	**8**	(3) An allele whose influence is more powerful than its partner
(d) Recessive allele	**6**	(4) A pair of alleles that carry different forms of the same gene
(e) Homozygous	**1**	(5) A sex cell
(f) Heterozygous	**4**	(6) An allele that is expressed only if the dominant form is absent
(g) Gamete	**5**	(7) First cell of the new organism
(h) Dominant allele	**3**	(8) The actual appearance produced by an organism's genes

2. Read the following genetic "rules" for monohybrid crosses. Beside each rule put a **T** or a **F** depending on whether you think it is true or false.

(a) Traits are controlled by a pair of genes. ***T***

(b) Recessive alleles mask dominant ones. ***F***

(c) Gametes have one allele for each trait. ***T***

(d) Different forms of a gene for the same trait are called alleles. ***T***

(e) The genotype describes the appearance. ***F***

(f) The phenotype lists the alleles present. ***F***

(g) In heterozygous individuals one allele is dominant and one is recessive. ***T***

(h) An individual displaying the dominant trait could have one of two genotypes. ***T***

4. In guinea-pigs, a black colour coat (B) is dominant to a brown colour coat (b).

(a) Write the genotype of a homozygous dominant individual. ***BB***

(b) Write the genotype of a heterozygous individual ***Bb***

(c) Write the genotype of a homozygous recessive individual ***bb***

(d) A heterozygous black male was mated with a homozygous brown female. Complete the punnet square to show the possible offspring genotypes.

	B	***b***
b	***Bb***	***bb***
b	***Bb***	***bb***

(e) What ratio of brown to black offspring is suggested by the

the completed punnet square. _____***1:1***_____

5. Humans either have the ability to roll their tongue (T) or they do not (t). Tongue rolling is dominant over non-tongue rolling. A married couple who are both tongue rollers have three children. One child is a non-tongue roller and the other two are tongue rollers.

(a) Draw a punnet square that shows how this could happen.

	T	***t***
T	***TT***	***Tt***
t	***Tt***	***tt***

(b) Is it possible for one of the children to be heterozygous?

Yes . Write this child's genotype. ___***Tt***___.

(c) What is the phenotype ratio derived from this punnet square? ***Tongue roller : non-tongue roller = 3:1***